THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 4: **Atmosphere and Climate** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** gravity	force of attraction
2	**(d)** troposphere	rain, fog, hail
3	**(a)** 80 to 700 km	temperature rises very high
4	**(a)** climate	30 years or more
5	**(b)** humidity	evaporation
6	**(a)** land breeze	land cools faster
7	**(b)** low pressure while the ocean has high pressure	winds blow in from the ocean
8	**(d)** October to December	post-monsoon
9	**(a)** weather forecasting	monitoring weather
10	**(c)** rainfall	in cm or mm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A mixture of gases around the Earth held by gravity (1); it shields us from radiation and regulates temperature (1).	1+1
12	Troposphere, stratosphere, mesosphere, thermosphere, exosphere (2).	2
13	Weather phenomena such as rain, fog and hail occur in it and most of the air and water vapour lie in it (2).	2
14	It filters harmful ultraviolet radiation from the Sun (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nitrogen 78%, oxygen 21% (1.5); argon, carbon dioxide, water vapour, neon, helium and ozone in small amounts (1.5).	1½+1½
16	Provides oxygen and carbon dioxide (1); protects from UV radiation and meteors (1); regulates temperature (1).	1+1+1
17	Troposphere: lowest layer with weather, up to about 12 km (1.5); stratosphere: up to 50 km, ozone layer and clear skies for flying (1.5).	1½+1½
	OR: Mesosphere: up to 80 km, temperature falls, meteors burn (1.5); thermosphere: 80–700 km, temperature rises, auroras (1.5).	1½+1½
18	Equator gets direct rays and is warmer (1.5); temperature drops with height in mountains (1.5).	1½+1½
	OR: Coastal areas have moderate climate and humidity (1.5); interior places have extreme temperatures (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Diagram of five layers (2). (ii) Troposphere: weather; stratosphere: ozone; mesosphere: meteors; thermosphere: auroras; exosphere: outermost (3).	2+3
	OR: (i) Upper troposphere or lower stratosphere (1). (ii) Free of weather disturbances (1). (iii) Troposphere (1). (iv) Ozone layer in the stratosphere (2).	1+1+1+2
20	(i) Short-term vs long-term (2). (ii) Temperature, humidity, precipitation, wind, pressure (1). (iii) Warmer near the equator; cooler with height (2).	2+1+2
	OR: (i) A (1). (ii) Monsoon (1). (iii) South-west monsoon winds (1). (iv) Rice and other water-loving crops (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 4: **Atmosphere and Climate** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** nitrogen	about 78%
2	**(b)** stratosphere	up to 50 km
3	**(b)** thermosphere	northern and southern lights
4	**(a)** insolation	heat source
5	**(a)** rain	precipitation
6	**(b)** seasonal reversal of wind direction	from Arabic mausim
7	**(a)** Brihatsamhita and Krishiparashara	positions of Moon and Sun
8	**(b)** June to September	main rains
9	**(c)** droughts	lack of rain
10	**(d)** atmospheric pressure	air pressure

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Weather is the state of the atmosphere at a place and time; climate is the average over a long period (1+1).	1+1
12	The incoming solar radiation intercepted by the Earth's surface (2).	2
13	Latitude, altitude, distance from the sea, cloud cover (1+1).	1+1
14	Water vapour in the air (1); from evaporation of water from land and water bodies (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Unequal heating gives pressure differences (1.5); air moves from high to low pressure as wind (1.5).	1½+1½
16	Day: land warmer, low pressure, so wind from sea (1.5); night: land cools faster, so wind from land to sea (1.5).	1½+1½
17	In summer land heats and a low-pressure area forms (1); the cooler Indian Ocean has high pressure (1); moist winds blow towards land and bring rain (1).	1+1+1
	OR: Sailors depended on seasonal winds (1.5); Arabs named the reversal ‘mausim’ meaning season (1.5).	1½+1½
18	By observing the Moon, Sun and nakshatras (1.5); texts like Krishiparashara and Brihatsamhita described methods (1.5).	1½+1½
	OR: December to February with low temperatures in the north (1); dry, clear weather (1); western disturbances bring rain to the northwest (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Pressure differences (1). (ii) Day: sea breeze; night: land breeze (2). (iii) Land heats and cools faster than the sea (2).	1+2+2
	OR: (i) Seasonal reversal of winds (1). (ii) Low pressure over land, high over the ocean; moist winds blow in (2). (iii) Heavy rain, farming, floods/droughts (2).	1+2+2
20	(i) Monsoon winds and western disturbances (1). (ii) Crop loss, houses damaged, displacement (1). (iii) Better drainage, embankments, early warnings (1). (iv) Forecasts and warnings (2).	1+1+1+2
	OR: (i) Winter, summer, monsoon, post-monsoon with features (3). (ii) Dec–Feb, Mar–May, Jun–Sep, Oct–Dec (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.